

Chap-01Fixed Point Representation

- Regular number (मानक संख्या)

The point is fixed

$$(10.215)_{10} \quad (110.1101)_2$$

General format:

$$x = \pm (\underbrace{d_1 d_2 d_3 \dots d_{k-1}}_{\text{Whole part}} \cdot \underbrace{d_k d_{k+1} \dots d_n}_{\text{Fraction part}})_\beta$$

digit

Base

Floating Point Representation

- F is a subset of all real numbers.

FCR

- Follows certain format:

$$x \in \mathbb{R} \Rightarrow f(x) \in F = \{ \pm (0.d_1 d_2 \dots d_n)_\beta \cdot \beta^e \mid d_i, \beta, e \in \mathbb{Z}; 0 \leq d_i \leq \beta-1; e_{\min} \leq e \leq e_{\max}; d_1 \neq 0 \}$$

Fixed point $\rightarrow (10.215)_{10} = (0.10215)_{10} \times 10^2$

fixed point $\rightarrow (110.1101)_2 = (0.1101101)_2 \times 2^3$

floating point Representation

General Format,

$$F = \pm (\underbrace{0.d_1 d_2 d_3 \dots d_m}_\beta)_\beta \cdot \beta^e$$

fraction/Mantissa/
Significant

Base

exponent $\begin{cases} \text{Integer} \\ e_{\max} \\ e_{\min} \end{cases}$

$[e_{\min}, e_{\max}]$

Convention 1: (General/Standard form)

$$F = \pm (0.d_1 d_2 d_3 \dots d_m)_\beta \cdot \beta^e$$

Where, $d_1 = 1$ [For other bases (Not binary)]
 $d_1 \neq 0$

Convention 2: (IEEE Normalize form)

$$F = \pm (0.1 d_1 d_2 d_3 \dots d_m)_\beta \cdot \beta^e$$

Convention 3: (IEEE Denormalize form)

$$F = \pm (1.d_1 d_2 d_3 \dots d_m)_\beta \cdot \beta^e$$

Q. $\beta=2, m=3, e=3$, highest value using convention 1, 2, 3?

Conv 1:

$$\pm (0.ddd)_2 \cdot 2^3$$

highest value: $(0.111)_2 \cdot 2^3$

$$= (1 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-3}) \times 2^3$$

$$= (0.875)_{10} \times 2^3 = (7)_{10}$$

Conv 2:

$$\pm (0.1ddd)_2 \cdot 2^3$$

highest value: $(0.1111)_2 \cdot 2^3$

$$= (7.5)_{10}$$

Conv 3:

$$\pm (1.ddd)_2 \cdot 2^3$$

highest value: $(1.111)_2 \cdot 2^3$

$$= (1 \times 2^0 + 1 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-3}) \cdot 2^3$$

$$= (15)_{10}$$

Lowest (Non negative)

Conv 1:

$$(0.100)_2 \cdot 2^3$$

$$= (4)_{10}$$

Conv 2:

$$(0.1000)_2 \cdot 2^3$$

$$= (4)_{10}$$

Conv 3:

$$(1.000)_2 \cdot 2^3$$

$$= (8)_{10}$$

lowest

Q. $\beta=2, m=3, e=[-1, 2]$

highest, lowest, lowest (Non negative) using Convention

1, 2, 3?

=>

Conv 1:

highest = $(0.111)_2 \cdot 2^2$

lowest = $(0.100)_2 \cdot 2^{-1}$
(Non neg)

lowest = $-(0.111)_2 \cdot 2^2$

Conv 2:

highest = $(0.1111)_2 \cdot 2^2$

lowest = $(0.1000)_2 \cdot 2^{-1}$
(Non neg)

lowest = $-(0.1111)_2 \cdot 2^2$

Conv 3:

highest = $(1.111)_2 \cdot 2^2$

lowest = $(1.000)_2 \cdot 2^{-1}$
(Non neg)

lowest = $-(1.111)_2 \cdot 2^2$

Q Base = 2, m = 3, e = [-1, 2], how many combination of number is possible using convention 1 and 2? (Non Negative)

Conv 1: $(0.d_1d_2d_3)_2 \times 2^e$

Base Digit = $2^e = 4$

$e = \{-1, 0, 1, 2\}$

Total combination = $4 \times 4 = 16$

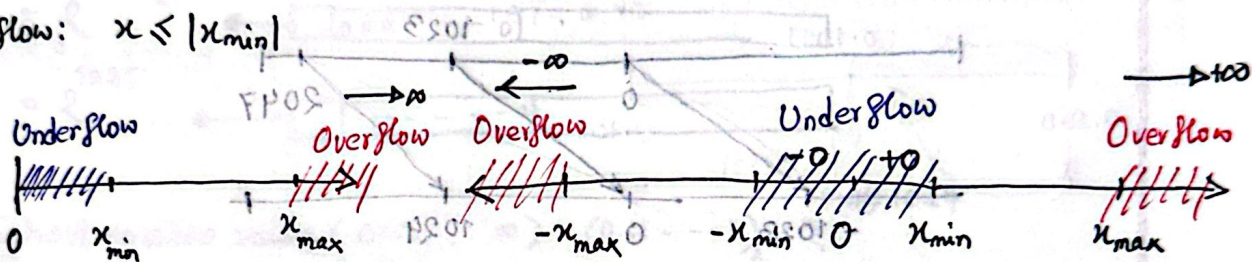
Conv 2: $(0.1d_1d_2d_3)_2 \times 2^e$

$1 = 2 \times (0.1) = \text{min}$

Total combination = $2^3 \times 4 = 32$

Q Overflow: $x \geq |x_{\max}|$

Underflow: $x \leq |x_{\min}|$



(Without neg)

(With neg)

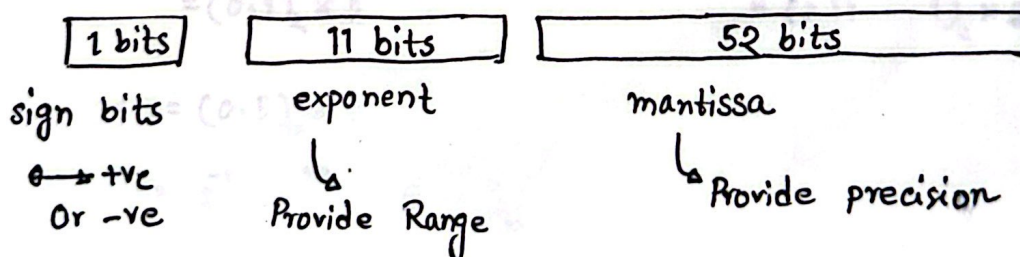
* Limitations of floating point representation

1. Range is finite causes Overflow & Underflow

2. Possible numbers are not equally spaced.

* IEEE Standard (1985) for double precision (64 bits)

64 bit system



Conv 3:

$$(1.d_1d_2 \dots d_m)_\beta \cdot \beta^e$$

11 bits $\rightarrow$ Total combination

$$= 2^{11} = 2048 \neq 2047$$

$$(1.d_1d_2 \dots d_{52})_2 \times 2^e$$

Range: $[0 - 2048]$

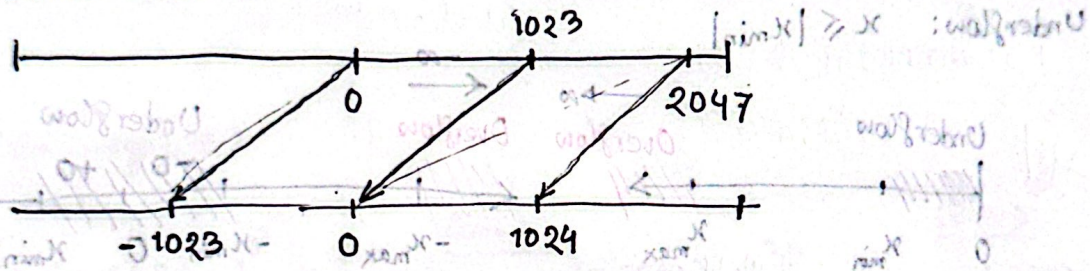
$$\text{Max} = (1.111 \dots 1)_2 \times 2^{2048} \rightarrow \text{too large value}$$

$\uparrow$ $e_{\min}$ $\uparrow$ $e_{\max}$

$$\text{Min (non neg)} = (1.0)_2 \times 2^0 = 1 \rightarrow \text{not that small}$$

Exponent Biasing:

Conv 3: IEEE DeNormalized



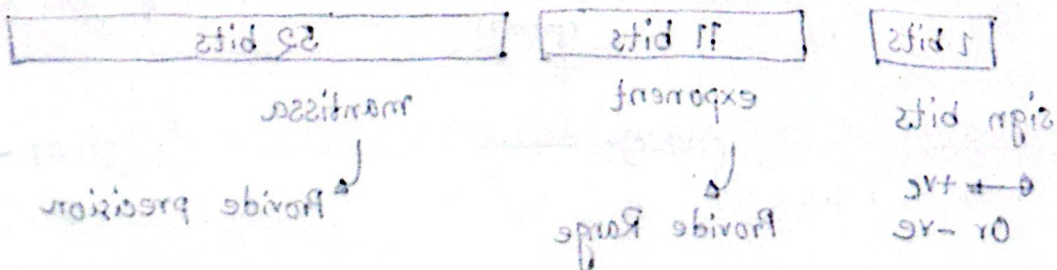
$$(1.d_1d_2 \dots d_{52})_2 \times 2^{e - \text{bias}}$$

$$\pm \infty = 2^{1024}$$

$$\Rightarrow (1.d_1d_2 \dots d_{52})_2 \times 2^{e - 1023}$$

$$\text{Max} = (1.111 \dots 1)_2 \times 2^{1024}$$

$$\text{Min (non neg)} = (1.0)_2 \times 2^{-1023}$$



Conv 2: IEEE Normalized

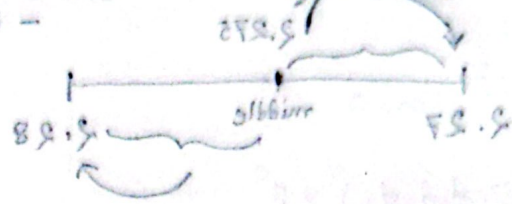
$$\pm (0.1d_1d_2 \dots d_{52})_2 \times 2^{e-1023}$$

$$\Rightarrow F = \pm (0.1d_1d_2 \dots d_{52})_2 \times 2^{e-1022}$$

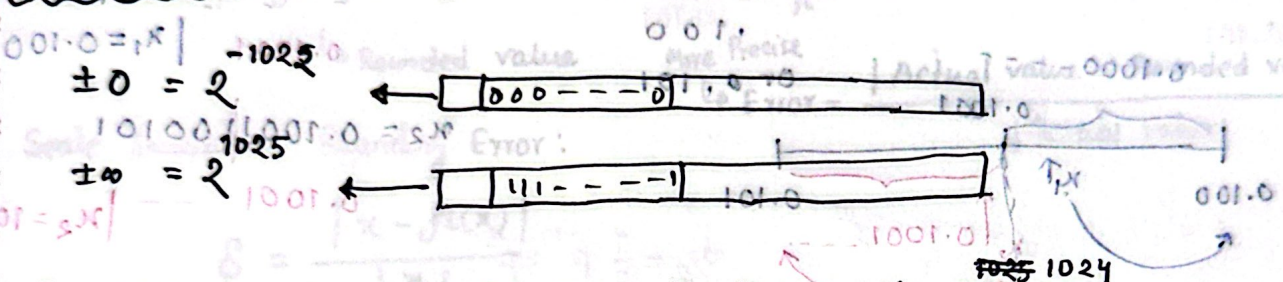
$2047 - 1022 = 2025$
 2025

$$\text{Max} = (0.111 \dots 1)_2 \times 2^{1025}$$

$$\text{Min (Non neg)} = (0.10)_2 \times 2^{-1022}$$



Special case



The highest possible value (except ∞) = $(0.111 \dots 1)_2 \times 2^{1024}$

The lowest " " " " (" 0) = $(0.100 \dots 0)_2 \times 2^{-1022}$

What is the lowest value possible (non negative) if bias is 512 in IEEE

Normalized 64 bit form?

$$\Rightarrow (0.1d_1d_2 \dots d_{52})_2 \times 2^{e-\text{bias}}$$

$$\text{lowest (non neg)} = (0.1)_2 \times 2^{0-512}$$

$$= (0.1)_2 \times 2^{-512}$$

$$= (0.5)_{10} \times 2^{-512}$$

$$= 2^{-1} \times 2^{-512} = (2^{-513})_{10}$$

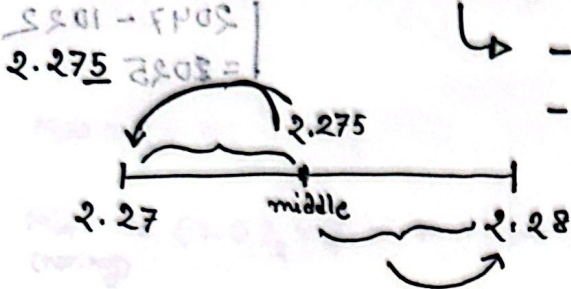
$$\text{Highest} = (0.111 \dots 1)_2 \times 2^{2047-512}$$

$$= (0.111 \dots 1)_2 \times 2^{1535}$$

Rounding

— (let's, my system 2 digit after decimal point)

$$2.276 \approx 2.28 \text{ (rounding)}$$

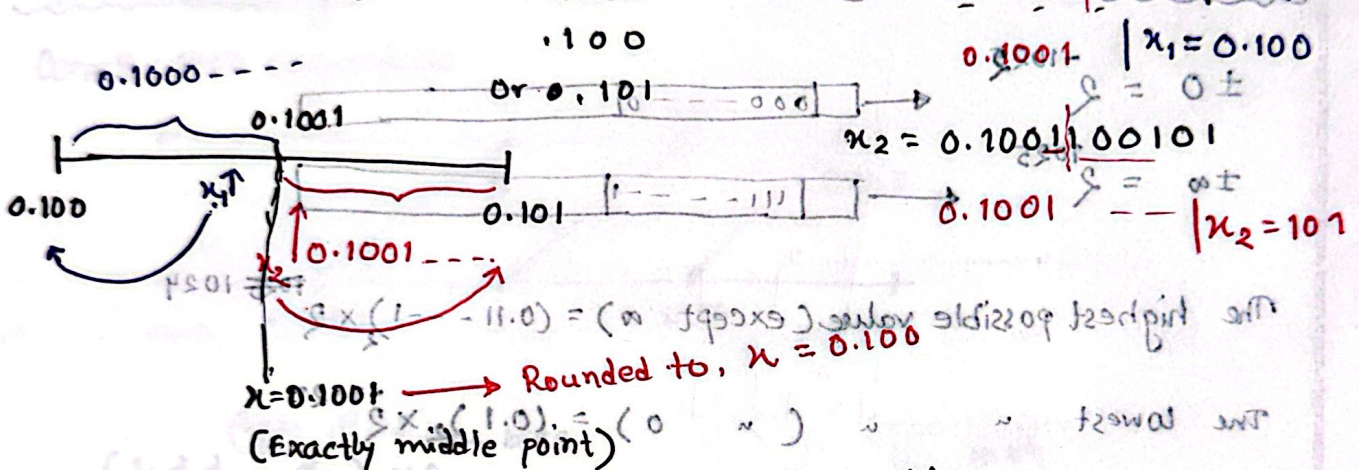


Under Count

Over Count

Now for BINARY:

let. mantissa $\rightarrow 3$



then Rounding: Closest middle point

0.100 ✓ (even)

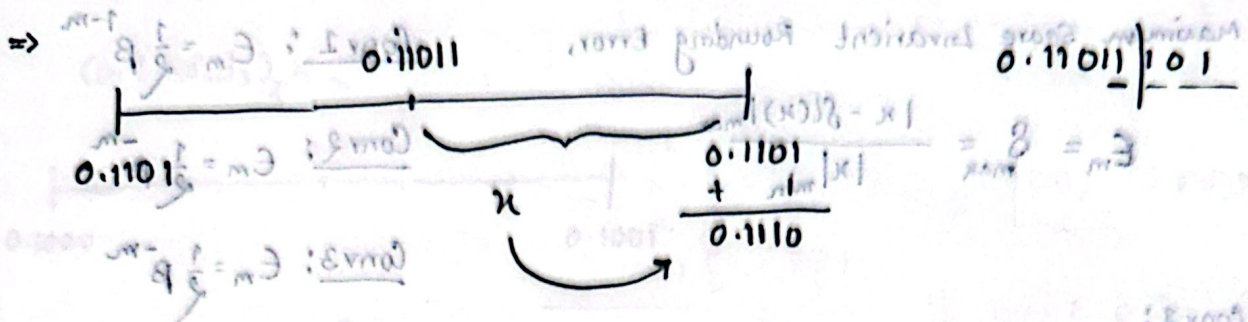
0.101

In BINARY:

* Ends with 0 $\rightarrow$ even

* Ends with 1 $\rightarrow$ odd

Q (0.1101101)₂ Round into m=4 system. (Machine epsilon)



$$\therefore (0.1101101)_2 \rightarrow (0.1101)_2$$

➡ Rounding Error

$$x = \text{---} \quad f(x) = \text{---}$$

↳ Rounded value

$$\text{Error} = \frac{|\text{Actual value} - \text{Rounded value}|}{|\text{Actual value}|}$$

More Precise
↳ Error =

Scale Invariant Rounding Error:

$$\delta = \frac{|x - f(x)|}{|x|}$$

$$\delta = \frac{|x - f(x)|}{|x|} = \frac{1}{2^m}$$

Q let. $x = (0.1101101)_2$ $m=4$ $\delta=?$

$$f(x) = (0.1101)_2$$

$$\delta = \frac{|(0.1101101)_2 - (0.1101)_2|}{|(0.1101101)_2|}$$

$$= \frac{|0.86328125 - 0.875|}{|0.86328125|}$$

$$= (0.0136)_{10}$$

$$= (0.0000001)_2$$

$$= (0.00000001101110110110)_2$$

$$0.1101 \quad 0.1101101$$

$$\delta = \frac{1}{2^m} = \frac{1}{2^4} = \frac{1}{16}$$

$$\begin{aligned} (0.0136)_{10} \times 2 &= 0 \\ (0.0272)_{10} \times 2 &= 0 \\ (0.0544)_{10} \times 2 &= 0 \\ (0.1088)_{10} \times 2 &= 0 \\ (0.2176)_{10} \times 2 &= 0 \end{aligned}$$

Machine Epsilon (ϵ_m) $\rightarrow \delta_{\max}$ $\rightarrow$ (10111011.0)

Maximum Scale Invariant Rounding Error,

Conv 1: $\epsilon_m = \frac{1}{2} \beta^{1-m}$

$$\epsilon_m = \delta_{\max} = \frac{|x - f(x)|_{\max}}{|x|_{\min}}$$

Conv 2: $\epsilon_m = \frac{1}{2} \beta^{-m}$

Conv 3: $\epsilon_m = \frac{1}{2} \beta^{-m}$

Conv 1: $m=3$

$(0.d_1d_2d_3)_\beta \times \beta^e \rightarrow (0.1)$

0.1001

0.101

0.100

$d/2$

$$\begin{aligned} d &= (0.101 - 0.100)_2 \times 2^e \\ &= (0.001)_2 \times 2^e \\ &= 2^{-3} \times 2^e \end{aligned}$$

$$d/2 = \frac{1}{2} \beta^{-m} \times \beta^e$$

$$\therefore |x - f(x)|_{\max} = \frac{1}{2} \beta^{-m} \times \beta^e$$

$d = \beta^{-m} \times \beta^e$

$|x|_{\min} = (0.d_1d_2d_3)_2 \times 2^e$

0.101

$(0.100)_2 \times 2^e$

$= 2^{-1} \times 2^e$

$|x|_{\min} = \beta^{-1} \times \beta^e$

$\epsilon_m = \frac{|x - f(x)|_{\max}}{|x|_{\min}}$

$\frac{1}{2} \beta^{-m} \times \beta^e$

$\frac{\beta^{-1} \times \beta^e}{\beta^{-1} \times \beta^e}$

$\frac{1}{2} \beta^{1-m} = \epsilon_m$

* Maximum error distance $= d/2$

$f(x) = x$

rounded value

Scale Invariant Rounding Error:

$(10111011.0) = x$

$(0.111.0) = f(x)$

$(0.111.0) - (0.111.0)$

$(0.111.0)$

$(0.111.0)$

$(0.111.0)$

$(0.111.0)$

$(0.111.0)$

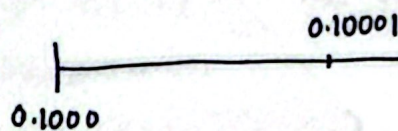
$(0.111.0)$

$(0.111.0)$

$(0.111.0)$

Conv 2: $m=3$

$$(0.1d_1d_2d_3)_2 \times 2^e$$



$$d = (0.0001)_2 \times 2^e$$

$$= 2^{-4} \times 2^e$$

$$|x - f(x)|_{\max} = \frac{1}{2} \beta^{-1-m} \times \beta^e$$

$$e_m = \frac{|x - f(x)|_{\max}}{|x|_{\min}}$$

$$= \frac{\frac{1}{2} \beta^{-m} \cdot \beta^{-1} \cdot \beta^e}{\beta^{-1} \times \beta^e}$$

$$e_m = \frac{1}{2} \beta^{-m}$$

*ShortCut:

From conv 1: $e_m = \frac{1}{2} \beta^{1-m}$

But in conv 2 (and) conv 3 the number of digit in mantissa is $m+1$

Therefore, $e_m = \frac{1}{2} \beta^{1-(m+1)} = \frac{1}{2} \beta^{-m}$

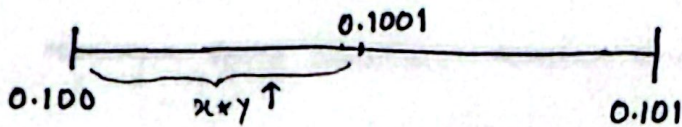
Q $\beta=2, m=3$; calculate rounding error for $(x * y)$, where $x = \frac{5}{8}, y = \frac{7}{8}$

$$x = \frac{5}{8} = \frac{1}{2} + \frac{1}{4} = 2^{-1} + 2^{-2} = (0.101)_2$$

$$y = \frac{7}{8} = \frac{1}{2} + \frac{2}{8} = 2^{-1} + 2^{-2} + 2^{-3} = (0.111)_2$$

$$x * y = \frac{5}{8} \times \frac{7}{8} = \frac{35}{64} = \frac{32}{64} + \frac{4}{64} = 2^{-1} + 2^{-5} = (0.100011)_2$$

$f(x \times y)$



$$f(x \times y) = (0.100)_2 = (2^{-1})_{10} = (\frac{1}{2})_{10}$$

$$\text{Rounding Error, } \delta = \frac{|\frac{35}{64} - \frac{1}{2}|}{|\frac{35}{64}|} = \frac{3}{35} = (0.08571428571)_{10}$$

Loss of Significance

$$f(x) = x + \delta_1 x, \quad f(y) = y + \delta_2 y$$

$$= x(1 + \delta_1) = y(1 + \delta_2) \quad \text{Error}$$

$(x \circ y) \rightarrow x$ composition y
(Algebraic $+, -, *, /$)

$$f(x \circ y) = (x \circ y) [1 + \delta_3]$$

$$= x(1 + \delta_1) \circ y(1 + \delta_2)$$

$$= (x \circ y) +$$

$$f(x \circ y) = f(x) \circ f(y)$$

$$= x(1 + \delta_1) \circ y(1 + \delta_2)$$

$$= (x \circ y) + (x \delta_1 \circ y \delta_2)$$

$$= (x \circ y) \left[1 + \frac{x \delta_1 \circ y \delta_2}{x \circ y} \right]$$

Now, δ_3 ,

For '-' operation,

$$\delta_3 = \frac{x \delta_1 - y \delta_2}{x - y}$$

Error significantly increases
When x closer to y

Scale invariant error.
 δ_3

if x and y are really close

then $x - y$ approaches to 0 (zero), which makes δ_3 (the error) larger.

δ_3 becomes ∞ when $x = y$.

$\therefore$ This phenomena is called Loss of Significance.

Example: Solve $x^2 - 56x + 1 = 0$ with 4 significant form.

* $P_n(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$

Let, $x_1, x_2, \dots, x_n$ are the roots

We can write,

$$(x-x_1)(x-x_2) \dots (x-x_n) = a_0 + a_1x + \dots + a_nx^n$$

Coefficient of x^0 : $x_0x_1x_2 \dots x_n = a_0$

Coefficient of x^1 : $x_2 \dots x_n + x_1x_3 \dots x_n + x_1x_2x_4 \dots x_n + \dots + x_1 \dots x_{n-1} = a_1$

Coefficient of x^n : $1 = a_n$

Solve $x^2 - 56x + 1 = 0$ with 4 significant form

$\Rightarrow$ Let, two roots are x_1 and x_2

$$(x-x_1)(x-x_2) = x^2 - 56x + 1$$

Coeff of x^0 : $x_1x_2 = 1$ — (i)

Coeff of x : $-x_1 - x_2 = -56$

$$\Rightarrow x_1 + x_2 = 56 \text{ — (ii)}$$

Coeff of x^2 : $1 = 1$

Using General solution,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = 28 \pm \sqrt{783} = 28 \pm 27.98$$

$$x_1 = 28 + 27.98 = 55.98 \checkmark$$

$$x_2 = 28 - 27.98 = 0.02000 \checkmark$$

lets check,

$$\textcircled{a} \rightarrow x_1 + x_2 = 55.98 + 0.02000$$

$$= 56.00 \text{ (Satisfies)}$$

$$\textcircled{b} \rightarrow x_1x_2 = (55.98)(0.02000)$$

$$= 1.120 \neq 1 \text{ (Do not satisfy)}$$

Now,

$$x_1x_2 = 1$$

$$\Rightarrow x_2 = \frac{1}{x_1} = \frac{1}{55.98}$$

$$x_2 = 0.01786$$

lets check again,

$$x_1 + x_2 = 55.98 + 0.01786$$

$$= 55.99786$$

$$= 56.00 \text{ (Satisfies)}$$

$$\therefore x_1 = 55.98, x_2 = 0.01786$$